

Consumer Shopping Behaviour Analysis

An end-to-end data analytics case study using Python, PostgreSQL, and Power BI

Python

PostgreSQL

Power BI

Executive Summary

3,900+Customer
Records**\$59.76**Avg. Purchase
Amount (USD)**3.75/5.0**Avg. Review
Rating**10**Business SQL
Queries

THREE-PHASE ANALYTICAL PIPELINE

01

Python EDA

Data ingestion, quality auditing, null imputation, feature engineering (age_group, purchase_frequency_days), SQLAlchemy load.

02

PostgreSQL

10 structured business queries using aggregations, correlated subqueries, CTEs, window functions, and conditional aggregation.

03

Power BI Dashboard

KPI cards, 5 interactive charts, 4 cross-filter slicers — stakeholder-ready, no SQL/Python required.

KEY FINDINGS

- 2.1x gender revenue gap — Male \$157,890 vs Female \$75,191 — unexplained by category mix alone
- 72% of repeat buyers (>5 purchases) hold no subscription — 2,518 pre-qualified conversion leads
- Near-50% discount rates on Hat, Sneakers, and Coat signal margin erosion risk requiring A/B testing

Presentation Agenda

01 Dataset & Objectives

3,900+ records · 6 analytical dimensions · dual objectives

02 Python EDA & Data Preparation

Null audit · imputation · feature engineering · SQLAlchemy load

03 PostgreSQL Business Queries

10 queries · revenue, products, discounts, loyalty, shipping

04 Revenue & Demographics Analysis

Gender gap · subscription impact · age group contribution

05 Product & Discount Insights

Top-rated products · discount concentration · basket analysis

06 Customer Loyalty & Segmentation

79% loyal base · 72% unsubscribed repeat buyers · segments

07 Power BI Dashboard

KPI cards · 5 charts · 4 slicers · self-serve analytics

08 Business Recommendations

6 prioritised actions · HIGH / MEDIUM / LOW impact

09 Conclusion & Deliverables

Technical proficiency · project files · extensibility



01

Dataset & Objectives

19 features · 3,900+ records · 6 analytical dimensions

Dataset Overview & Feature Set

3,900+

Total Records

19

Feature Columns

2

Engineered Features

1

Null Column (review_rating)

FEATURE CATEGORIES

Customer Demographics

age, gender, location, age_group (engineered)

Transaction Details

purchase_amount, payment_method, discount_applied, shipping_type, season

Product Information

item_purchased, category, color, size, review_rating

Loyalty & Behaviour

subscription_status, previous_purchases, frequency_of_purchases, purchase_frequency_days (eng.)

Note: promo_code_used dropped — 100% identical to discount_applied (confirmed via Boolean equality check).

02

Python EDA & Data Preparation

Ingestion · Cleaning · Feature Engineering · Database Load

Data Cleaning & Feature Engineering

Step 01 | Initial Audit

`df.info()`, `.describe()` confirmed $3,900 \times 18$ shape. Only `review_rating` had nulls.

Step 02 | Null Imputation

Category-median imputation on `review_rating` — preserves group-level distributions without distorting the overall mean.

Step 03 | Column Standardisation

All headers lowercased, whitespace stripped, spaces → underscores.
`purchase_amount_(usd)` renamed.

Step 04 | Redundant Feature Removal

`promo_code_used` confirmed 100% duplicate of `discount_applied` using `.all()` check. Dropped.

ENGINEERED FEATURES

age_group

`pd.qcut(df['age'], q=4)` — Quantile-based banding ensures equal-frequency groups, making inter-group comparisons statistically balanced.

Young Adult

Adult

Middle-aged

Senior

purchase_frequency_days

Maps categorical frequency labels to integer day counts, enabling numerical analysis and future predictive modelling.

Weekly → 7d

Bi-Weekly → 14d

Monthly → 30d

Quarterly → 90d

Annually → 365d

03

PostgreSQL Business Querying

10 queries · 7 SQL techniques · pgAdmin 4

10 Business Queries — Techniques & Coverage

Q1 Revenue by Gender

SUM / GROUP BY

Q3 Top 5 Products by Rating

AVG / ORDER BY / LIMIT

Q5 Subscription vs. Revenue

COUNT / SUM / AVG

Q7 Customer Segmentation

CTE + CASE WHEN

Q9 Repeat Buyers & Subscriptions

WHERE / GROUP BY / COUNT

Q2 Discounters Above Avg Spend

Correlated Subquery

Q4 Avg Spend by Shipping Type

ROUND / AVG / WHERE

Q6 Discount Rate by Product

CASE WHEN / Cond. Agg.

Q8 Top Products per Category

CTE + ROW_NUMBER()

Q10 Revenue by Age Group

SUM / GROUP BY / ORDER BY



Revenue



Discount



Product



Shipping



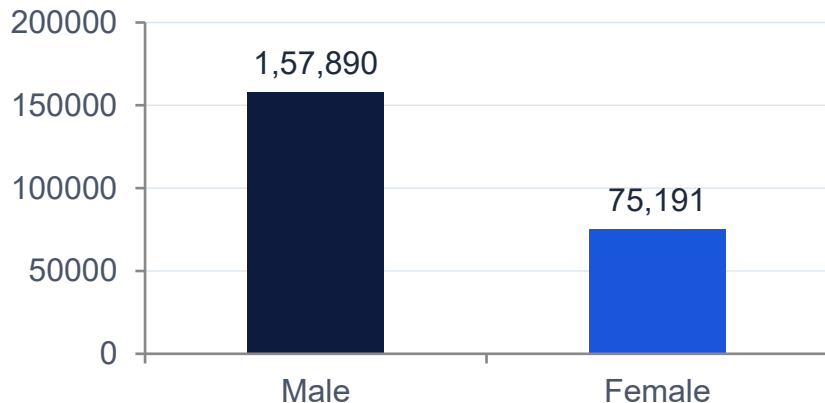
Loyalty



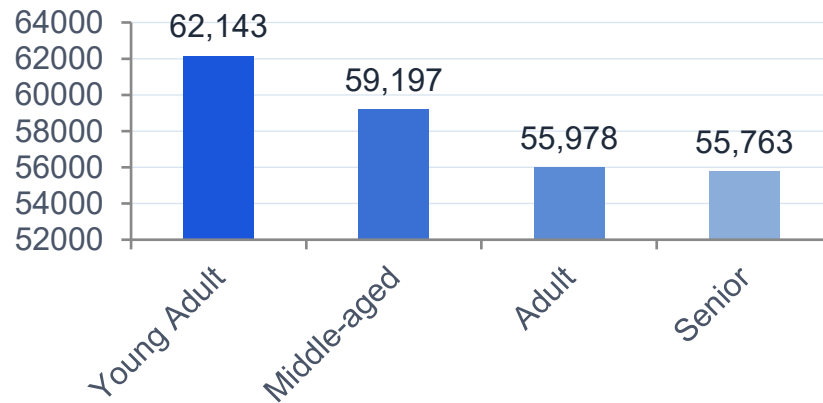
Demographics

Revenue Analysis

Q1 — Revenue by Gender



Q10 — Revenue by Age Group



KEY INSIGHT

2.1× Gender Gap

Male revenue (\$157,890) is 2.1× female (\$75,191). In clothing/accessories — where female demand typically exceeds male — this 52% revenue shortfall points to acquisition funnel underrepresentation or product-segment misalignment.

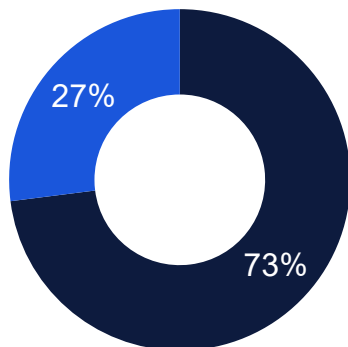
KEY INSIGHT

Young Adults Lead

Young Adults generated \$62,143 — the highest of four age bands. Notably all four groups fall within a \$6,380 range, confirming broad cross-generational product appeal. Young Adult lead is notable given price-sensitivity of this cohort.

Subscription Impact Analysis

Q5 — Subscription Status



■ Non-Subscriber ■ Subscriber

2,847

Non-Subscribers

73% of customer base

\$0.38

Spend Difference

Subscribers vs Non-subscribers per transaction

2,518

Repeat Non-Subscribers

72% of >5-purchase customers hold no subscription

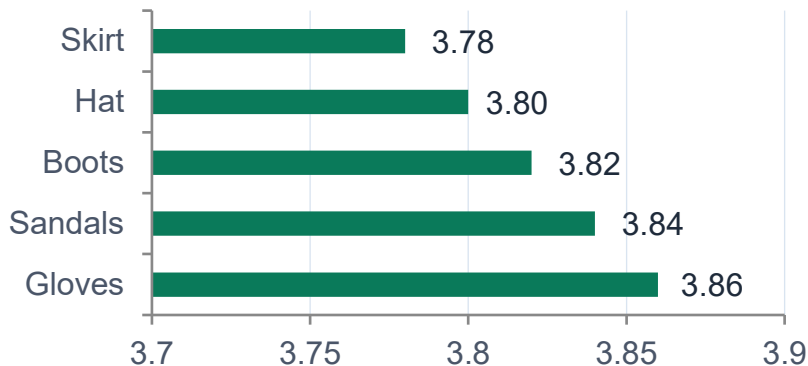
STRATEGIC IMPLICATION

Subscription does not drive higher per-visit spend (\$0.38 difference). Its value is retention & frequency. Target the 2,518 pre-qualified repeat buyers with exclusive benefits (free shipping, early access, loyalty points).

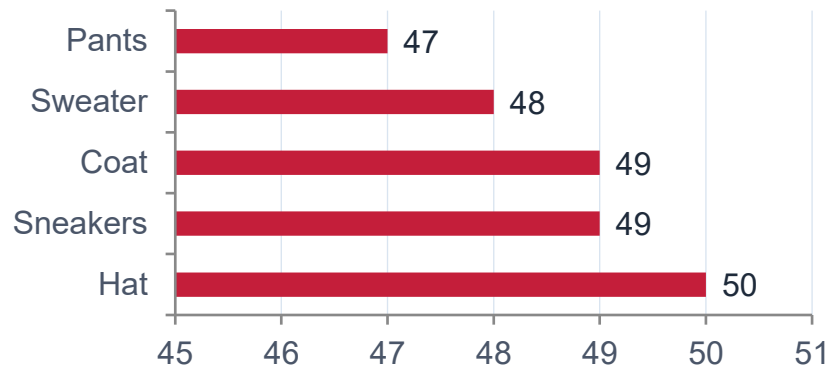
Q9 Finding: Among customers with >5 purchases, 72% (2,518) hold no subscription — the highest-priority conversion audience in the entire dataset.

Product Performance & Discount Analysis

Q3 — Top 5 Products by Avg Rating



Q6 — Top 5 Products by Discount Rate



Quality Leaders Are Accessories & Footwear

Top 5 rated items (Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78) — narrow range signals consistently high satisfaction. Prioritise in homepage hero & email campaigns.

839 Discount Users Spent Above Average

Q2: 839 customers applied a discount and still spent $\geq \$59.76$ (dataset average). Discounts function as conversion accelerators, not margin eroders — but near-50% rates on 5 SKUs need A/B testing.

Basket Expansion Opportunity

Q8: Jewelry (171 orders) leads Accessories; Blouse & Pants (171 each) lead Clothing. Bundling high-volume accessories with top clothing items is an untested basket-expansion lever.

Customer Loyalty & Segmentation

LOYAL

79%
3,116

customers · > 10 purchases

RETURNING

18%
701

customers · 2–10 purchases

NEW

2%
83

customers · 1 purchase only

Q4 — Shipping Behaviour Insight

Express shipping customers spend \$2.02 more on average (\$60.48 vs \$58.46 Standard). This self-selection effect — willingness to pay for speed correlates with higher basket values — supports a checkout-stage upsell nudge offering Express shipping above a minimum basket threshold (e.g., 'Upgrade to Express for orders over \$75').

ACQUISITION RISK

Only 83 customers (2%) are classified as New. While a 79% Loyal base is commercially strong, extreme dependence on existing customers without new acquisition inflow creates long-term growth risk.

A new customer acquisition strategy is as critical as loyalty retention — the current base will naturally attrite without replenishment.

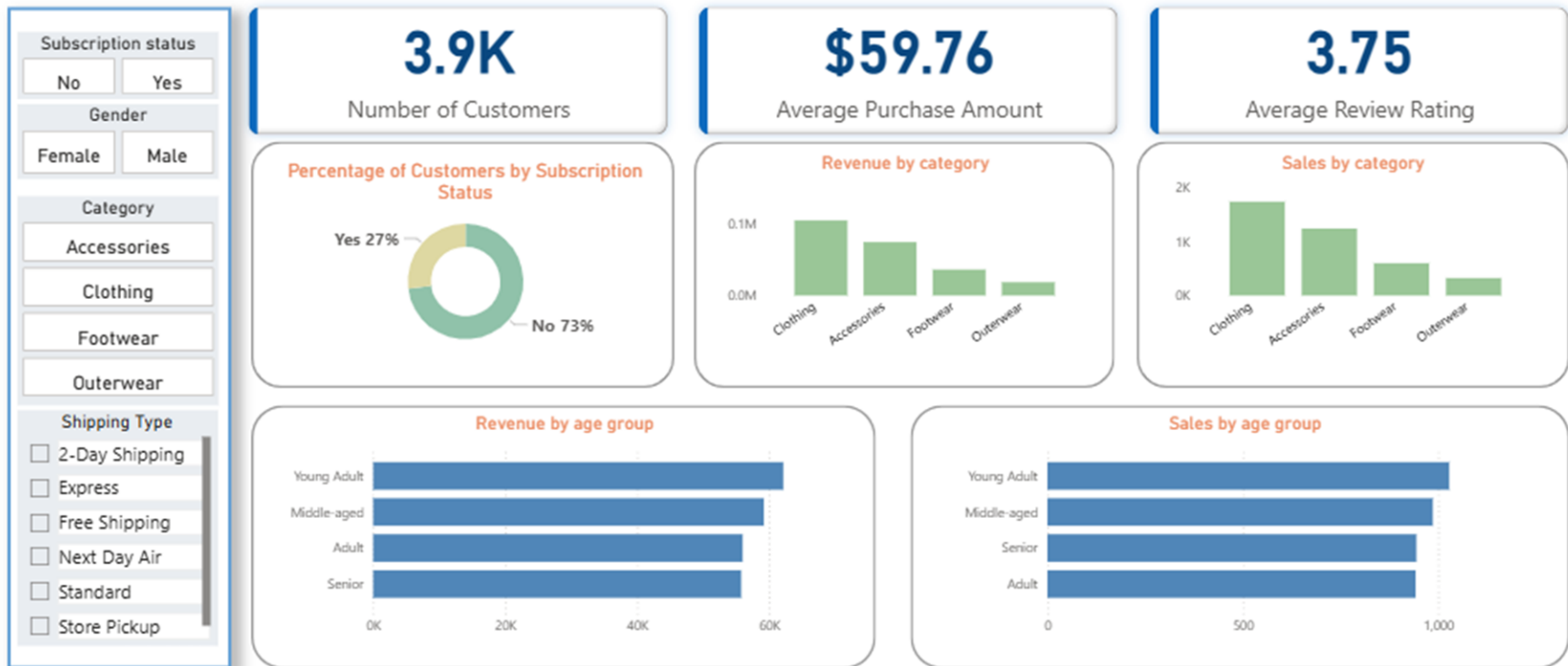
04

Power BI Dashboard

Dashboard · KPI Cards · 5 Charts · 4 Cross-Filter Slicers · Self-Serve Analytics

Interactive Dashboard Architecture

Customer Shopping Behaviour Dashboard



Interactive Dashboard Architecture

KPI SUMMARY CARDS

3.9K

Number of Customers

\$59.76

Avg. Purchase Amount

3.75

Avg. Review Rating (/ 5.0)

CHART COMPONENTS

Subscription Status Split

Doughnut Chart

73% non-subscriber vs 27% subscriber — loyalty penetration gap

Revenue by Age Group

Horizontal Bar

Young Adults lead (\$62K); all groups within \$6.4K range

Revenue by Category

Vertical Bar

Clothing leads; Outerwear trails — guides category investment

Sales Volume by Age Group

Horizontal Bar

Young Adults lead order count; Senior slightly below Adult

Sales Volume by Category

Vertical Bar

Validates demand concentration; consistent with revenue ranking

INTERACTIVE SLICERS

- Subscription Status (Yes/No)
- Gender (Female/Male)
- Category (4 options)
- Shipping Type (6 options)



05

Business Recommendations

6 Prioritised Actions Directly Traceable to Query Outputs

6 Data-Backed Business Recommendations

HIGH

#1 Loyalty → Subscription

Finding: 72% of >5-purchase customers (2,518) are non-subscribers

Launch targeted subscription conversion campaign with exclusive benefits: free shipping thresholds, early sale access, loyalty points.

MED

#4 Young Adult Retention

Finding: Young Adults are the top revenue-generating segment at \$62,143

Strengthen retention via mobile-first UX, early access to sale events, and personalised re-engagement journeys.

HIGH

#2 Discount Optimisation

Finding: Hat (50%), Sneakers & Coat (49%) carry near-50% discount rates

A/B test reduced discount depths on top-frequency items. Monitor conversion rate at 20%, 30%, and current levels.

LOW

#5 Express Shipping Upsell

Finding: Express buyers spend \$2.02 more on average than Standard buyers

Add checkout-stage upsell nudge tied to minimum basket value (e.g., 'Upgrade to Express for orders over \$75').

MED

#3 Female Acquisition

Finding: Female revenue (\$75,191) is 52% lower than male (\$157,890)

Invest in gender-targeted digital campaigns, female-skewed influencer partnerships, and personalised recommendation engines.

LOW

#6 Top-Rated Amplification

Finding: Gloves (3.86), Sandals (3.84), Boots (3.82) lead on review rating

Feature in homepage hero sections, email headers, and as anchor items in cross-category product bundles.

06

Conclusion & Project Deliverables

End-to-End Analytical Pipeline · Reproducible · Extensible

Conclusion & Project Deliverables

THREE MOST COMMERCIALY SIGNIFICANT FINDINGS

1

72% Repeat Buyers Are Non-Subscribers

2,518 pre-qualified, high-intent customers have demonstrated loyalty without any subscription incentive — the highest-ROI conversion target in the dataset.

2

2.1× Unexplained Gender Revenue Gap

Male (\$157,890) vs Female (\$75,191) in a product mix where female demand typically dominates — signals acquisition funnel or personalisation failure.

3

Near-50% Discount Rates on 5 Major SKUs

Hat, Sneakers, Coat, Sweater, and Pants each show 47–50% discount rates. Full-price purchasing is no longer the norm — margin erosion risk is immediate.

TECHNICAL PROFICIENCY DEMONSTRATED

Python

Pandas · SQLAlchemy · pd.qcut · group-median imputation

PostgreSQL

Aggregation · Correlated Subqueries · CTEs · Window Functions · Conditional Aggregation

Power BI

KPI Cards · Multi-chart Dashboards · Cross-filter Slicers

PROJECT DELIVERABLES

- .ipynb Python Notebook — EDA, cleaning, feature engineering, SQLAlchemy load
- .sql PostgreSQL Script — 10 annotated business queries with full output comments
- .pbix Power BI Dashboard — Interactive KPIs, 5 charts, 4 cross-filter slicers
- .docx Project Report — Full methodology, findings, recommendations, and tables

Thank You

Consumer Shopping Behaviour Analysis
